

Undercover Assistance: Designing a Disguised App to Navigate Sexual Harassment

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ABSTRACT

Sexual harassment remains a critical safety concern for many women. Despite extensive discussions within the HCI community on technological solutions to address harassment, prior research has primarily focused on cyber harassment, with relatively fewer studies addressing design interventions to assist women *during* incidents of sexual harassment in physical spaces. To this end, we conducted a survey with 106 women who have experienced sexual harassment, aiming to identify common contexts of occurrence and the coping strategies employed. The survey revealed that social gatherings, such as parties, are the most frequent settings for sexual harassment. Coping strategies varied based on the relationship with the assailant; only 6% of respondents would call 911 when harassed by an acquaintance, compared to 19% when harassed by a stranger. The most common strategy involved pretending to have an urgent matter and leaving the scene, followed by contacting friends for assistance. However, many participants expressed fear of escalating the situation by provoking the perpetrator. Drawing from these insights, we propose the design of a disguised app that mimics a social media app, thus reducing suspicion while using an app in close proximity to the perpetrator. The app includes multiple functionalities recommended by survey participants: 1) triggering a fake call from a male voice, 2) sharing location and discreetly chatting with friends, 3) recording, and 4) an emergency alarm. This paper discusses how these design features address the challenges of safely navigating sexual harassment incidents.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing systems and tools**; *Human computer interaction (HCI)*;

KEYWORDS

sexual harassment, safety app, wearable technology

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1 INTRODUCTION

Sexual harassment, defined as unwelcome sexual advances and other verbal or physical conduct of a sexual nature, is a critical safety concern for women [8]. A 2018 national study conducted by the UC San Diego Center on Gender Equity and Stop Street Harassment (SSH) reported that over 80% of women in the United States have experienced sexual harassment in their lifetime, underscoring its extensive prevalence [12]. In the field of Human-Computer Interaction (HCI), a growing body of research has explored design interventions to address sexual harassment [16]. However, in our examination of 96 works published for the last 5 years in major HCI communities, including CHI, CSCW, UIST, DIS, and IDC, we observed that 71% of the works focused on sexual harassment in online spaces. These works primarily addressed issues such as abuse on social media [2, 11, 13, 14, 17, 19], consent on dating apps [4, 10, 22], and automated detection of abusive language and behaviors online [18, 21]. Relatively few studies have addressed harassment in physical spaces.

Among the few studies that have examined physical sexual harassment "*in situ*," the proposed solutions have primarily included showing crime rates using crowdsourced maps [1, 9, 20] and notifying emergency contacts [1]. While these previous works provide valuable insights, they predominantly focus on a single setting—public places. To the best of our knowledge, no prior HCI work has comprehensively examined how preferred safety measures differ depending on the context and "during" the moment of sexual harassment. For example, victims' responses are known to differ when harassment occurs by acquaintances (e.g., friends, co-workers) in intimate spaces compared to strangers in open public spaces [5, 15]. In the former case, victims are often hesitant to seek formal help, such as calling 911, and are less likely to take assertive measures due to fear of escalation or damaging existing relationships [15].

To address these issues, this study has two primary goals: (1) to understand preferred safety measures in different contexts and (2) to suggest design interventions that can help women safely navigate various sexual harassment situations. Our work makes several contributions to the existing HCI literature. First, our survey with over a hundred sexual harassment victims examines the most desired safety features for future mobile devices in varying safety incidents, thus providing designers with insights to prioritize their design efforts. Second, we propose unique design features based on respondents' feedback, including a "disguised" mobile and wearable app to seek help from others. Our survey indicated that contacting friends is among the most common responses, yet fear of provoking perpetrators remains a primary concern. Disguised features can help potential victims avoid suspicion from perpetrators and safely ask for help, considering that sexual harassment often occurs at close distances where perpetrators can see how victims interact with their mobile apps.

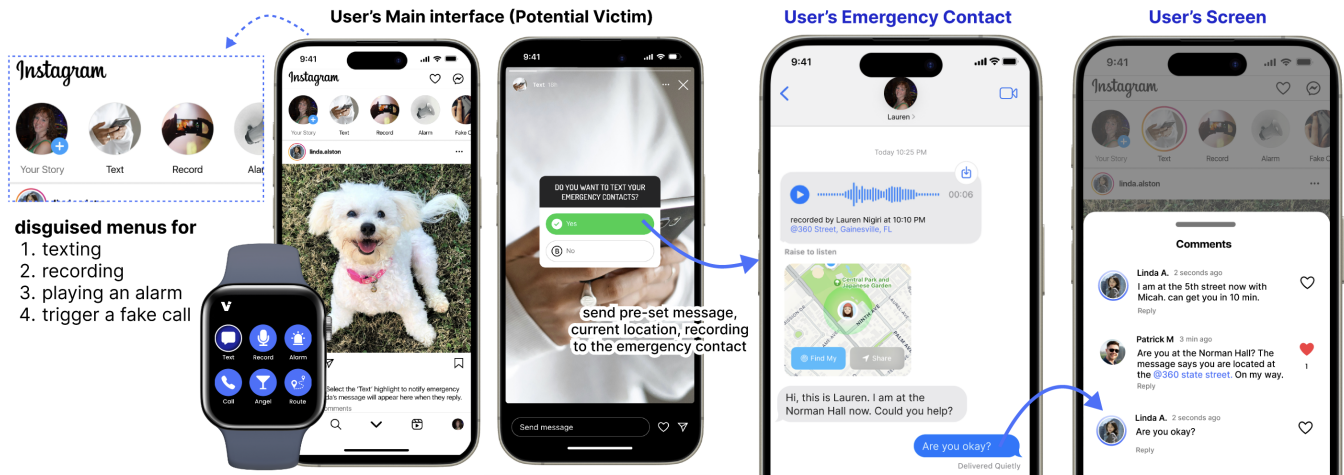


Figure 1: Screenshot of the app’s user interface. Users can activate the ‘discrete’ mode, which mimics popular social media’s interface. The top profile images act as a discreet menu, offering options for texting, recording, and triggering a fake phone call. Clicking the “text” button sends a preset message along with user’s location information to an emergency contact.

2 SURVEY ON SEXUAL HARASSMENT

Our survey focused specifically on women in the United States to narrow the scope, as harassment towards women is more common than towards men [7, 12]. Respondents were recruited through the online survey platform. To target women who have experienced sexual harassment, we included a screening question. A total of 106 respondents indicated they had experienced unwanted physical contact and sexual comments. The average age of the respondents was 32 years old. The survey asked about the common contexts of sexual harassment, coping strategies, current safety measures, and preferred safety features on mobile devices. The survey results indicated that most common places were social gatherings such as parties (55%), followed by workplaces (49%), public spaces like parks and transportation (49%), and bars and restaurants (45%). The most common perpetrators were strangers (75%) and acquaintances (45%), followed by coworkers (41%) and family members (10%). The survey also inquired about coping strategies when the harassment was committed by strangers versus acquaintances. When harassed by a stranger, the most common coping response was “asking for help if other people are around” (58%), whereas in the latter case, “pretending to have an urgent matter and leaving” was the most common strategy (56%). Regarding current safety measures, 49% of respondents indicated that they use mobile apps, primarily featuring location sharing. Respondents also rated various safety features on a Likert scale (1: not important at all, 5: very important). Among seven items, the highest-rated features included sending a quick pre-set message to friends ($M = 4.57$), location sharing ($M = 4.36$), recording ($M = 4.35$), calling 911 ($M = 4.22$), wearable watch sync ($M = 4.09$), and a fake call from a male voice ($M = 3.86$). However, in the open-ended questions, several respondents expressed concerns about the effectiveness of these safety measures:

“I am a little worried that the popularity of the app could make it less effective since the harasser will know about it when I use it.”

“Opening this kind of app that says ‘fake call’ will make it pretty obvious to the person harassing me.”

3 VI: UNDERCOVER SAFETY APP

Based on the survey data, we designed a mobile app called Vi. Among many design features, we discuss two main interface below.

Disguised App to Avoid Suspicion: Survey respondents raised concerns about the effectiveness of safety app due to the close physical proximity in incidents of sexual harassment. Perpetrators could notice that victims are using a safety app, potentially escalating the situation. To address this, we proposed a disguised app that mimics the look of a social media app, as illustrated in Figure 1. We specifically chose Instagram based on feedback in our iterative design process, given its popularity among young adults.

Communicating with Emergency Contacts: Our survey results indicated that communicating with friends via pre-set messages is the most desirable safety measure. Vi is designed to send the pre-set message, location information, and recording to the emergency contact with a single button press. Constant communication is also crucial during the incident, as victims need to know how long it will take for help to arrive and describe their situation. To facilitate this, Vi allows users to communicate with their emergency contact via an interface that mimics the comments section of an Instagram post. This design makes it less obvious to the harasser that the potential victim is texting their emergency contact.

4 EVALUATION AND FUTURE WORK

To evaluate the usability and perceived effectiveness of the Vi mobile app, we conducted user testing with 11 participants. Each participant was asked to complete a series of tasks using the Vi prototype and then complete a survey. The survey included questions on the System Usability Scale (SUS) [3], perceived effectiveness of the app, and intention to use the app [6]. All participants indicated that they are likely to use the app in the future, and 88% agreed that the app is effective in addressing sexual harassment. Future studies will conduct a more comprehensive evaluation to examine the usefulness of various design features in different contexts.

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